



CikguSync

An Offline-First CPD Companion for Rural Sabah Teachers

Digital and Hybrid Teacher Professional Development in Rural Sabah, Malaysia

Assignment 3

ICT for the Global South
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Group 3

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09 May 2026

1. Overview of the Proposed Solution

1.1 Problem and core idea

Assignment 2 showed that rural teachers in Sabah face a specific and well-documented gap in professional development. Teachers at disconnected schools cannot reliably access online CPD, there is no portable way to record classroom application, and district and national offices are losing their main evidence source as SPLKPM shuts down in 2026 (Utusan Malaysia, 2025). These problems are not new: weak infrastructure, heavy workloads, and professional isolation have been consistently reported among rural Sabah teachers (Jaafar et al., 2025), and existing platforms like DELIMa 2.0 were built for connectivity conditions that most Interior schools simply lack.

CikguSync is our proposed response to that gap. The name draws directly from the context: '*cikgu*' is the everyday Malay word for teacher, and '*sync*' describes what the platform does technically, storing CPD progress and classroom evidence on the device first and uploading it whenever a connection becomes available. *CikguSync* is built as a progressive web app that teachers can install or pin on onto their phones. Its basic workflow is: download a small CPD module bundle during a morning 4G window or at a NADI or PKG centre, complete the module offline at any point, capture a low-resolution photo as classroom evidence, tag it to an SGM 2.0 domain, and let the app sync the record later when connectivity returns.

The logo on the title page reflects the same ideas. An open book sits at the centre as a symbol of professional learning. Circular arrows around it represent the offline-first sync cycle. The diamond pattern along the book's spine is inspired by *linangkit*, a traditional embroidery style from Sabah's Lotud and Rungus communities, which grounds the design in Sabah specifically rather than in generic edtech imagery. The colour palette follows the navy used in Malaysian Ministry of Education materials and stays legible as a phone home-screen icon.

CikguSync is not meant to replace DELIMa 2.0, PPG 2.0, Program Duta Guru, UNICEF FS4A, or MDEC's #CikguJuaraDigital. Those programmes are already embedded in the system and represent substantial investment. *CikguSync* sits alongside them as a lightweight, offline-tolerant layer that can eventually exchange structured records with these systems once institutional agreements are in place.

1.2 Target users and value proposition

Three personas from Assignment 2 anchor the design. *Cikgu Liana*, a primary teacher in Nabawan, uses offline modules to learn practical techniques, photographs classroom evidence, and syncs her record when a connection is available. *Guru Besar Jainal*, a school leader in Pitas, receives a weekly school summary by Telegram or email rather than being expected to manage another platform. *Pegawai Norhaida*, a PPD district officer in Keningau, can access an aggregated dashboard

that includes data from offline schools and exports records for SSH and DPD reporting.

2. Preconditions and Interoperability

2.1 Preconditions for use

CikguSync is designed for low-resource conditions, but it still rests on a few basic assumptions. Teachers need access to a personal or school Android phone capable of running a modern browser. Continuous internet is not required, but occasional connectivity is needed to download modules and sync records. This can happen during a morning 4G window, at a NADI or PKG centre, at home on weekends, or during a district visit. On the institutional side, BPG and PPD would need to approve the CPD module structure and its mapping to SGM 2.0 domains. School leaders need a simple accepted channel for receiving summaries, such as Telegram or email, and district officers need at least one stable online location for reviewing synced data and exporting reports.

The solution deliberately avoids requirements that would be unrealistic in most Interior schools. It does not depend on high-speed internet, continuous cloud access, devices for every pupil, or a school-side administrator. The prototype also does not require native app store deployment. Leaving these out was a deliberate choice based on what the context realistically allows.

2.2 Social interoperability

Social interoperability means the solution fits the governance structures and everyday routines already in place around it. *CikguSync* positions BPG as the owner of professional development and uses SGM 2.0 as the recognition language for teacher growth (MOE, 2023), so it does not introduce a new competency framework that teachers and administrators would need to learn separately. It translates what teachers already do into evidence that counts within an existing national standard.

At school level, the *Guru Besar* receives useful summaries without taking on a technical administrator role. At the district level, PPD and JPN officers use an aggregated dashboard to see which schools are progressing, which remain effectively offline, and where support visits or NADI/PKG sessions should be directed next.

2.3 Technical interoperability

The prototype is standalone but built to be integration-ready. Data is stored using SGM 2.0 domain codes together with teacher and school identifiers, so records can later be exported or connected to DELIMa 2.0, Program Duta Guru, UNICEF FS4A, or BPG/JPN reporting systems without restructuring the data model. The system exposes simple REST endpoints and supports CSV and JSON export. Full integration is not promised at the prototype stage because it would require formal agreements with MOE units and partner organisations, which are outside the scope of a student project.

3. High-Level Technical Design

3.1 Main concepts and class diagram

The system has four main concept groups: users and organisations, learning objects, evidence and sync, and reporting. The UML class diagram in Figure 1 shows the static data structure. In the model, a Teacher completes CPDModule items through ModuleProgress, creates EvidenceSubmission items, and maps each submission to an SGMDomain. This lets the system connect teacher activity to SGM 2.0.

When the device is offline, EvidenceSubmission items are held in the SyncQueue. A weekly summary then aggregates evidence at the school level for the school leader to view.

Three design decisions in the model are important.

First, EvidenceSubmission and SyncQueue are kept as separate entities. This is better than using one record with a status field because low-connectivity uploads may be retried several times. If the same evidence is uploaded twice, the API can use evidenceId to accept only one submission while still keeping retry information in the sync queue for troubleshooting.

Second, ModuleProgress is separated from EvidenceSubmission because completing a module and capturing classroom evidence may happen at different times. A teacher might finish a module today and only collect classroom evidence two weeks later. These actions need different timestamps and different meanings in the audit trail.

Third, SGMDomain is an explicit class with a domainId and an indicatorCode, rather than as a hard-coded string field. If the SGM 2.0 framework changes in the future, or if an additional UNESCO ICT-CFT mapping is added (UNESCO, 2018), the framework can be updated as data rather than through a major schema change.

3.2 User interaction flow

The main user flow in Figure 2 begins with the teacher, since Option 1 focuses on the rural teacher CPD loop. The teacher opens the PWA, downloads a module during a connectivity window, completes it offline, captures evidence, maps it to an SGM 2.0 domain, and syncs the submission later.

The flow is linear. Each step represents a teacher action that must work without an internet connection, while synchronisation is delayed until connectivity becomes available. This matches teachers' experience of the system: a sequence of ordinary teaching and CPD activities, with the network used only when available.

The flow is also suitable for demonstration. The network can be turned off while the teacher completes the module and captures evidence, then turned back on to show how the queued submission syncs automatically.

3.3 System architecture

The architecture, as shown in Figure 3, has a teacher PWA, an intermittent connectivity layer, a Supabase prototype backend, and two lightweight reading views. The teacher PWA contains an offline cache, local IndexedDB storage, evidence capture, and a sync worker. The backend contains authentication, a relational database, object storage for compressed evidence images, REST API endpoints, and row-level security rules. School leaders and district officers read summaries from the same backend rather than maintaining separate spreadsheets, so reporting reflects a single source of truth.

3.4 Offline submission and failure handling

Offline operation is the main requirement, not an extra feature. Figure 4 shows the control flow. When a teacher submits evidence offline, the app stores the image and metadata locally, adds the item to the sync queue, checks connectivity from time to time, and uploads only when a connection is available. Sync must be idempotent, so repeated uploads of the same item do not create duplicate records in the dashboard.

The design handles three common failure cases.

First, if the network drops during upload, the item returns to the queue. The retry counter increases and the app waits longer before trying again.

Second, if the item still fails after a set limit, for example ten attempts over 14 days, it is marked as failed. The teacher then sees a clear "tap to retry" option, so the record is not silently lost.

Third, pupil privacy is protected through face blurring. The first version uses on-device face detection to blur faces before the image enters the queue. A server-side check runs after sync. This process should not block the teacher unless the image is unusable. The goal is to reduce risk while keeping the workflow practical in classrooms with low light, crowded rooms, or older phones.

Multi-device conflict handling is out of scope for the first release because the design assumes one teacher uses one main device.

4. Technology Infrastructure and Design Choices

4.1 Mobile vs web-based

CikguSync is built as a progressive web app, a website that behaves like an app. Teachers can open it in a browser, save it to their Android home screen, use it offline, and get updates automatically without ever visiting an app store. That matters because downloading and updating a traditional app requires things many rural teachers don't always have: a stable connection, a Google account, and a device that can handle the process smoothly.

4.2 Online vs offline functionality

The system is built to work offline first. Teachers can open modules they've already downloaded, complete

activities, write reflections, capture evidence, and save their submissions, all without needing a connection. Android Chrome can quietly delete stored data when the phone runs low on space or when a user clears their browser, so local storage isn't treated as permanent. Think of it more like a working copy that can be topped up the next time a teacher reaches an NADI or PKG centre. A connection is only needed for three things: downloading new modules, syncing saved records, and refreshing the dashboard. This was a deliberate decision shaped by the reality on the ground in rural Sabah. When the school router cuts out, rain breaks the 4G signal, or a teacher only gets online for a few minutes a day, the system still needs to do its job. With somewhere between 300 and 400 schools in Sabah still without internet access (The Star, 2025), building around connectivity would mean failing the schools that need this the most.

4.3 Cloud vs local hosting

Supabase is a realistic backend for the student prototype because it gives you authentication, Postgres tables, object storage, REST APIs, and row-level security, and it does so quickly. For a real-world public sector deployment, we propose to migrate the backend to MOE/BPG-controlled hosting or an approved Malaysian government cloud so that long-term ownership and control of teacher data is retained by the Malaysian education system. Supabase is useful for demonstration purposes; production data residency and ownership decisions would be made by BPG in accordance with MOE policy.

4.4 Integration, voice, and text

The first release is standalone but integration-ready. Full integration with DELIMa 2.0 or partner systems is postponed because it requires institutional access and API agreements. The data model, however, already uses SGM 2.0 domains and exportable records. The core interface is text-first because text is easier to translate, compress, cache, and search. Optional short audio summaries can support lower-literacy users or parent-facing explanations, but audio is not the main interaction mode because it increases data use, and Sabah's linguistic diversity makes reliable speech-to-text impractical without a custom acoustic model.

5. Human Aspects

5.1 Digital literacy and familiar behaviour

The solution assumes teachers are comfortable with basic phone actions: opening WhatsApp, taking a photo, downloading a file, and checking a status icon. It does not assume any confidence with cloud tools or education platforms beyond that.

The teacher interface uses plain labels such as "Download module", "Use offline", "Add evidence", "Saved on this phone", and "Sync when online". The school-leader summary is designed to feel like the WhatsApp reporting routines most school leaders already use. Both choices reduce the number of new behaviours required and lower the barrier to first use.

5.2 Language accessibility

The launch interface is available in Bahasa Malaysia and English. Bahasa Malaysia is the default because it is the national language of instruction. English covers CPD content and technical terms where Bahasa Malaysia equivalents are less established.

Kadazandusun and Iban are supported as addable language packs for selected module summaries rather than full-system translations in the first release. Sabah's linguistic diversity is real, but translating the entire platform into all local languages is not achievable in a prototype. The first batch of Kadazandusun summaries would be co-authored with #CikguJuaraDigital Champion Teachers from the Interior Division (Sabah Media, 2024), so the content reflects how teachers in that area actually speak rather than a generic translation.

5.3 Accessibility, inclusion, and workload

Rural Sabah primary teachers are predominantly women, and the design takes this into account. A teacher with childcare or family responsibilities at home often cannot stay at school late to use an evening 4G window. The morning-download model fits her daily routine better than a sync that requires her to be at school after hours. Data costs are also a practical constraint: a 40 GB Celcom SIM in Sabah costs around RM35, and a single live webinar can exhaust the entire allowance in one session. Keeping module bundles under 25 MB and relying on image-based evidence means monthly CPD data use stays well below 200 MB, so professional development does not compete with family communication or pupils' homework support.

For older teachers or those less confident with digital tools, the same plain labels that help new users also help here: "Download module", "Use offline", "Add evidence", "Saved on this phone", "Sync when online". These are paired with high contrast, large tap targets, and clear sync status indicators. The interface meets WCAG 2.1 AA contrast and tap-target standards. Optional audio summaries support teachers with low vision and parents with limited literacy, and no essential information is conveyed through colour alone.

Pupils are indirect beneficiaries rather than direct users. The data rules in Section 6 protect them: faces are blurred before upload, names are discouraged in reflections, and the first release has no pupil-facing component. Vulnerable pupils, including orphans, children of refugees from across the Indonesian border, and children with disabilities, are not made more visible to the platform than any of their peers.

Rural teachers are also at risk of being marginalised by measurement systems built around online activity, which tend to count connected schools far more readily than disconnected ones. CikguSync records offline activity and syncs it later, so a teacher working at a school without internet is not logged as inactive simply because of where she was posted.

6. Data Management, Privacy, and Accountability

6.1 Data collected

CikguSync collects only data needed for CPD access, evidence logging, syncing, and reporting. It does not collect pupil names, precise GPS coordinates, private family data, or unrelated teacher performance data. The main data groups are:

- **Teacher profile:** name, school, role, and language for account and CPD record purposes.
- **Module progress:** started/completed status and downloaded modules for learning continuity.
- **Evidence metadata:** SGM domain, module, date, and short reflection for CPD recognition.
- **Evidence image:** low resolution classroom photo with faces blurred before upload.
- **Sync metadata:** queued time, synced time, retry count, and audit status for reliability.
- **Aggregated dashboard data:** school and district counts for support planning and reporting.

6.2 Storage, access, and control

To support offline use, the records are temporarily stored on the teacher's device using browser storage (e.g., IndexedDB). When the internet is available, the sync worker uploads the records to the backend. The evidence images are stored separately from the metadata and are compressed. The device blurs faces before upload so no identifiable images of pupils should leave the school.

Access is role-based. Teachers can see their own modules, evidence and CPD record. Guru Besar see school level summaries and limited teacher level progress if needed for support. PPD officers can see district data and summaries of schools. BPG and JPN get compiled reports and exports. Only once sharing has been approved, external partners can see records relevant to their programme.

Data ownership should remain with MOE/BPG and the Malaysian public education system. Technical providers should act as processors under public-sector rules and contract limits, in line with the Personal Data Protection (Amendment) Act 2024.

6.3 Misuse prevention, transparency, and accountability

- **Data minimisation:** collect only what is needed for CPD recognition and support.
- **On-device anonymisation:** blur pupil faces before upload and discourage pupil names in reflections.
- **Role-based access:** apply row-level security so users see only appropriate records.
- **Audit logs:** record when data is created, changed, viewed, exported, or deleted.
- **Aggregated reporting:** use school and district summaries where individual data is not required.
- **Retention rules:** keep evidence for a defined CPD cycle, then archive or delete according to MOE policy.

Transparency. A context-aware transparency feature is built into the teacher interface. Each evidence item shows a simple "Who can see this?" label, for example *Only me*, *My school leader*, *PPD officer*, or *Aggregated for BPG*

reporting. After syncing, the teacher receives a data-use receipt showing what was uploaded, when it was synced, which SGM domain it supports, and who can access it.

Accountability. Three accountability mechanisms answer the question "who is responsible when something goes wrong?" First, audit logs are reviewed quarterly by a designated PPD data officer, with anomalies (mass exports, unusual access patterns) flagged automatically. Second, an anonymised summary of what data was collected and which partners read it is published quarterly on the JPN Sabah website and circulated through PIBG WhatsApp groups, modelled on UNICEF FS4A reporting practices (UNICEF Malaysia, 2024). third, each PPD designates a serving teacher, not an administrator, as a data ombudsperson. The ombudsperson receives complaints from peers and escalates them through JPN Sabah. This recognises the asymmetry between an individual rural teacher and a national platform, and adapts a governance pattern already familiar from community-level Native Court roles.

7. Updated Requirements

7.1 Requirements retained and changed

After moving deeper into the solution domain, the core requirements remain valid: downloadable CPD micro-modules, offline completion, image-based evidence, queue-and-sync, SGM 2.0 mapping, a portable CPD record, low data use, Android compatibility, privacy protection, and basic school and district views. The main change is scope reduction. Assignment 3 selects Option 1 as the first release and keeps mentoring only as a later or lightweight extension.

- **Teacher CPD:** keep as the main product loop and prototype focus.
- **Evidence:** keep image-based classroom evidence, with on-device face blurring and low-resolution upload.
- **Sync:** keep queue-and-sync as the main demo scenario.
- **Mentoring** (the focus of ContextSquaRE 2 in Assignment 2): deferred. The data model supports it through later extension of evidence submissions, but the prototype concentrates on the teacher CPD loop from ContextSquaRE 1.
- **School-leader view:** keep minimal weekly summary, with no daily login expectation.
- **District dashboard:** keep simple SGM 2.0 dashboard and export; postpone full partner integration.
- **Languages:** launch Bahasa Malaysia and English; add Kadazandusun summaries for selected modules later.
- **Integration:** prototype is standalone but API and export-ready; full integrations with DELIMa, Duta Guru, and FS4A are postponed.

7.2 MoSCoW priority for first release

Must have: offline PWA shell, downloadable CPD modules, image evidence capture, SGM 2.0 mapping, sync queue, idempotent sync, teacher CPD record,

Android support, data minimisation, and on-device face blurring.

Should have: school weekly summary, district dashboard, Bahasa Malaysia and English interface, REST and CSV export, accessibility basics, audit log, and weekly digest delivery by Telegram bot or email. Telegram is chosen over WhatsApp because its Bot API is free, server-driven, and already familiar to teachers in the Interior who use Telegram groups for lesson sharing (Persona 1, Assignment 2).

Could have: Kadazandusun summaries for selected modules, mentor check-in log, and partner-programme tags.

Won't have in this release: pupil-facing client, full external-system integration, full translation into all Sabah languages, live video training, or any function that requires continuous school internet.

8. Conclusion

CikguSync is a small ICT solution for teacher professional development in rural Sabah. The technical design is based on the local conditions of connectivity, governance structures of schools, language diversity and existing Malaysian education programs, rather than assuming technology alone solves the problem. The offline-first CPD loop focus means the solution works directly for Cikgu Liana but still gives Jainal dan Norhaida the summaries they need. The result is realistic to prototype, treats sensitive school data responsibly, and fits well enough in the wider education ecosystem to be extended after the first release.

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Appendix A: Figures

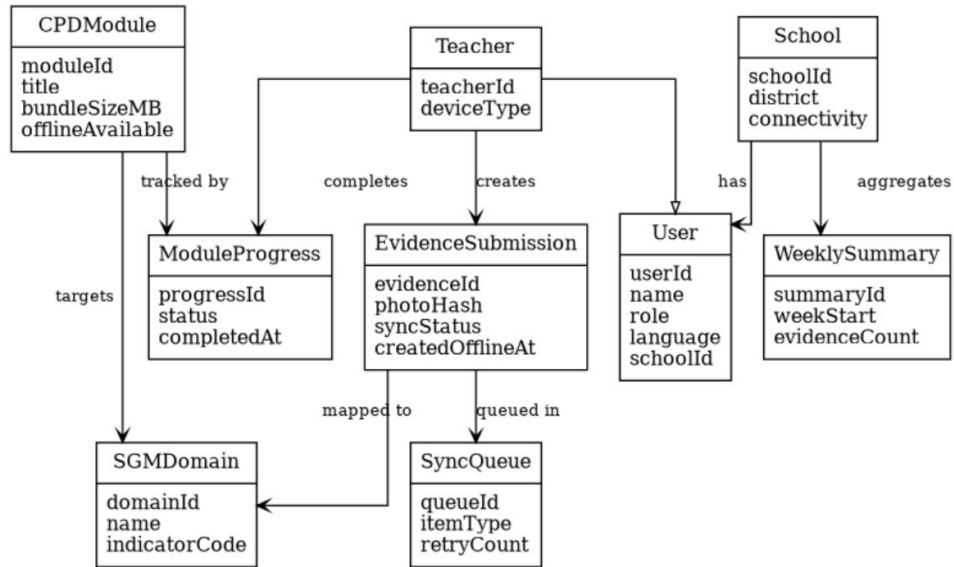


Figure 1. UML class diagram showing the main static data structures of CikguSync. EvidenceSubmission and SyncQueue are separated to enable idempotent retries; SGMDomain is modelled as an explicit class so future framework changes can be introduced as data.

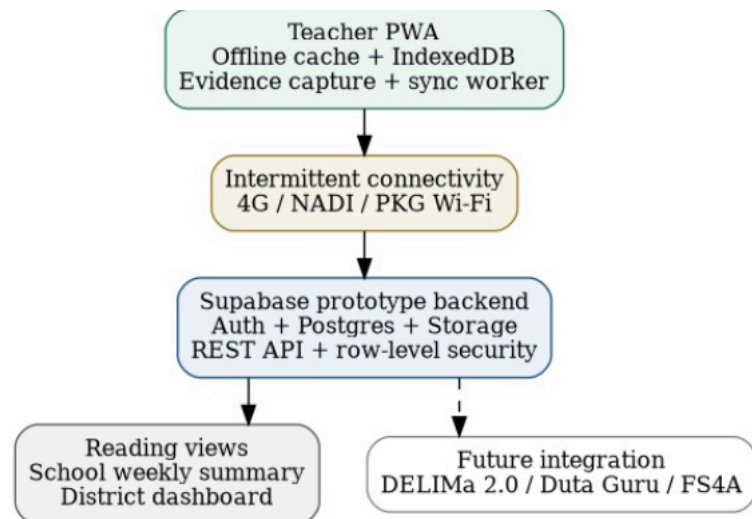


Figure 3. High-level system architecture for prototype and future integration. The teacher PWA, intermittent connectivity layer, Supabase backend, and reading views form four loosely coupled tiers.

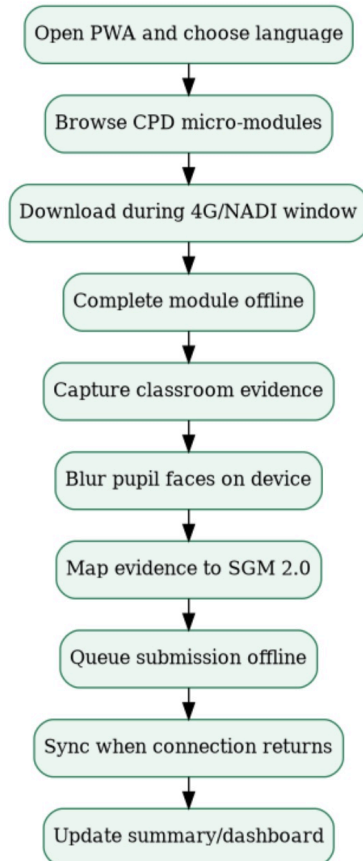


Figure 2. Teacher user flow from module download to evidence sync. The flow is intentionally linear: each step works independently, with synchronisation deferred until a window opens.

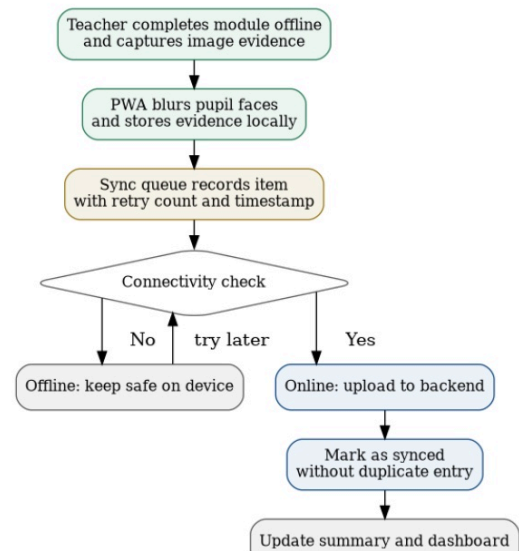


Figure 4. Control-flow sequence for offline evidence submission and later sync. The connectivity check is the only branching point; failure paths return the item to the queue with a retry counter.